Parental Vaccine Hesitancy and Its Evidence Base: A Short Review

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TEST DRAFT constructed for the VerRes HackPSU demo. Not a real manuscript; problems are planted on purpose.

Abstract

Vaccine hesitancy remains a public-health concern. This short review summarizes the evidence parents encounter and how the literature has evolved over time.

1. Introduction

Concerns about childhood vaccination grew after a 1998 case series reported an association between the measles, mumps and rubella vaccine and developmental disorders [1].

Large epidemiological studies have since examined this question, and a meta-analysis of case-control and cohort studies found that vaccines are not associated with autism [2].

Deep neural networks are increasingly used to analyse public-health text at scale [3].

2. Discussion

Some authors still cite the original case series as evidence of a vaccine-autism link [1], and the meta-analysis is sometimes cited as showing a weak positive association [2].

Communication strategies should present the full evidence base to parents.

References

[1] Wakefield, A. J., Murch, S. H., Anthony, A., et al. (1998). Ileal-lymphoid-nodular hyperplasia, non-specific colitis, and pervasive developmental disorder in children. The Lancet, 351(9103), 637-641. doi:10.1016/S0140-6736(97)11096-0

[2] Taylor, L. E., Swerdfeger, A. L., & Eslick, G. D. (2014). Vaccines are not associated with autism: an evidence-based meta-analysis of case-control and cohort studies. Vaccine, 32(29), 3623-3629. doi:10.1016/j.vaccine.2014.04.085

[3] LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. Nature, 521(7553), 436-444. doi:10.1038/nature14539